

Homelab Infrastructure Documentation: Proxmox Host

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Intent for this project

I built this homelab to have a real, hands-on environment for cybersecurity and IT administration practice, rather than just studying concepts in isolation. It doubles as a portfolio piece since employers in defensive security and sysadmin roles care a lot about seeing actual infrastructure work, not just certifications or coursework. Finally, I believe it'll be a genuinely good project for learning, honing skills learned throughout University, and experimenting with avenues I enjoy.

This project started July 4, 2026, and is actively evolving. This document gets updated as new pieces get added.

Why Proxmox

I chose Proxmox VE over alternatives like ESXi or running everything in Docker on bare Linux for two main reasons. First, it is free and open source, so there is no licensing cost or restriction getting in the way of experimenting. Second, it gives me real flexibility between full VMs and lightweight LXC containers on the same host, which mirrors the kind of virtualization environment I would expect to work with in an actual IT admin or sysadmin role. Learning Proxmox specifically is also directly relevant experience for that kind of job. Furthermore, prior academic experience with Proxmox allowed for an accelerated deployment timeline.

Why not use dedicated hardware?

The machine hosting all of this is a repurposed office PC rather than something bought specifically for homelabbing. Two reasons factored into that decision. First, it kept the cost of the project at basically zero since the hardware was already sitting unused. Second, it forces me to actually think about resource constraints, since I am working with 16GB of RAM total rather than an unlimited budget of compute. Having to plan around that, deciding what runs full time versus on demand, is its own kind of practical systems administration skill.

Total hardware cost: \$0. The machine was already sitting unused, repurposed instead of bought.

Physical Setup

The machine sits next to my main desktop. It runs headless, no monitor, mouse, or keyboard plugged in, since everything is managed through the Proxmox web UI over the local network.

Hardware specifications

CPU	Intel Core i5-6600 @ 3.30GHz, 4 cores, 1 socket
RAM	15.54 GiB total
Storage	1TB external SSD
Swap	8.00 GiB
Network interfaces	enp0s31f6 (onboard NIC), enx5065f34d1da3 (added USB NIC)
Boot mode	EFI
Hypervisor	Proxmox VE, pve-manager 9.2.2
Kernel	Linux 7.0.2-6-pve

At the time of writing, the host is running comfortably under load: 0.76% CPU usage across 4 cores, 10.29% RAM usage (1.60 GiB of 15.54 GiB), 4.57% disk usage on the root filesystem, and 0% swap usage. With this in mind, In the future I'll need to use VMs like Wazuh & other tools on demand and not powered on 24/7.

Having a second NIC already in place (the USB adapter) is worth noting since it sets up the future pfSense VM, which needs its own dedicated interface separate from the one currently handling general LAN traffic.

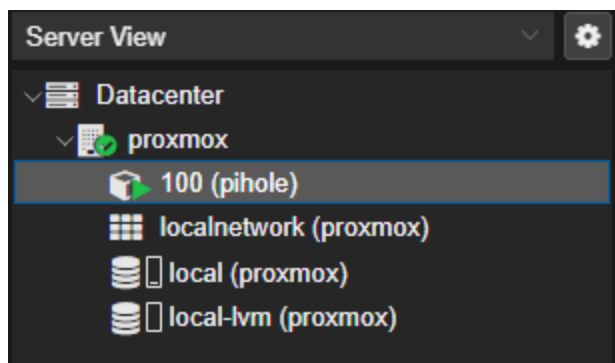
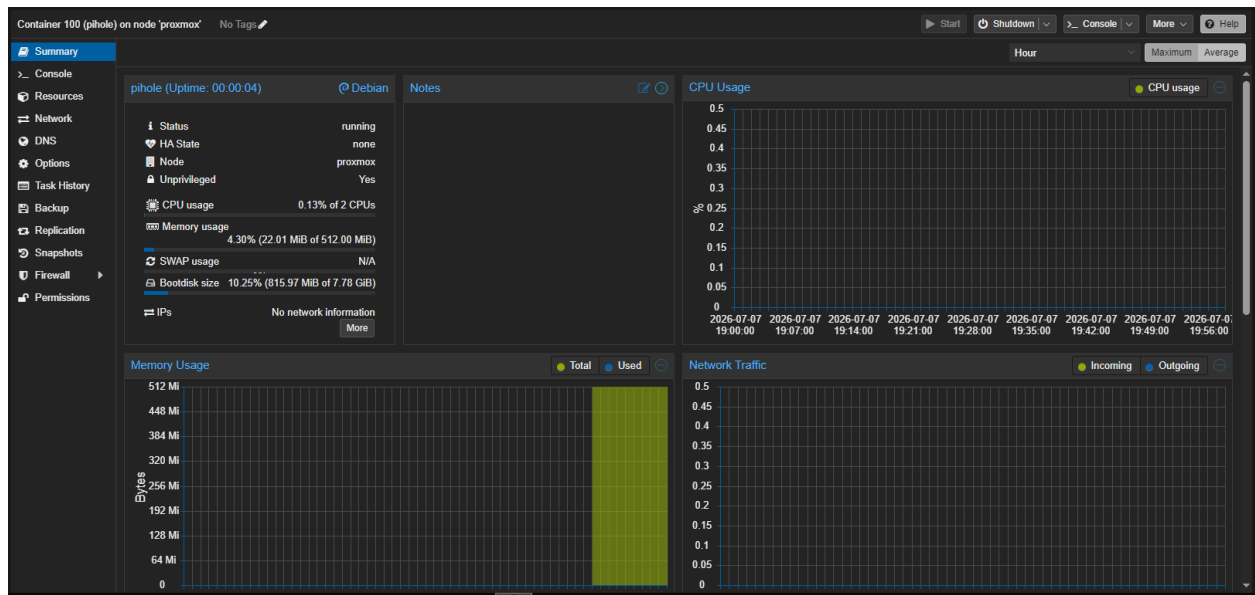
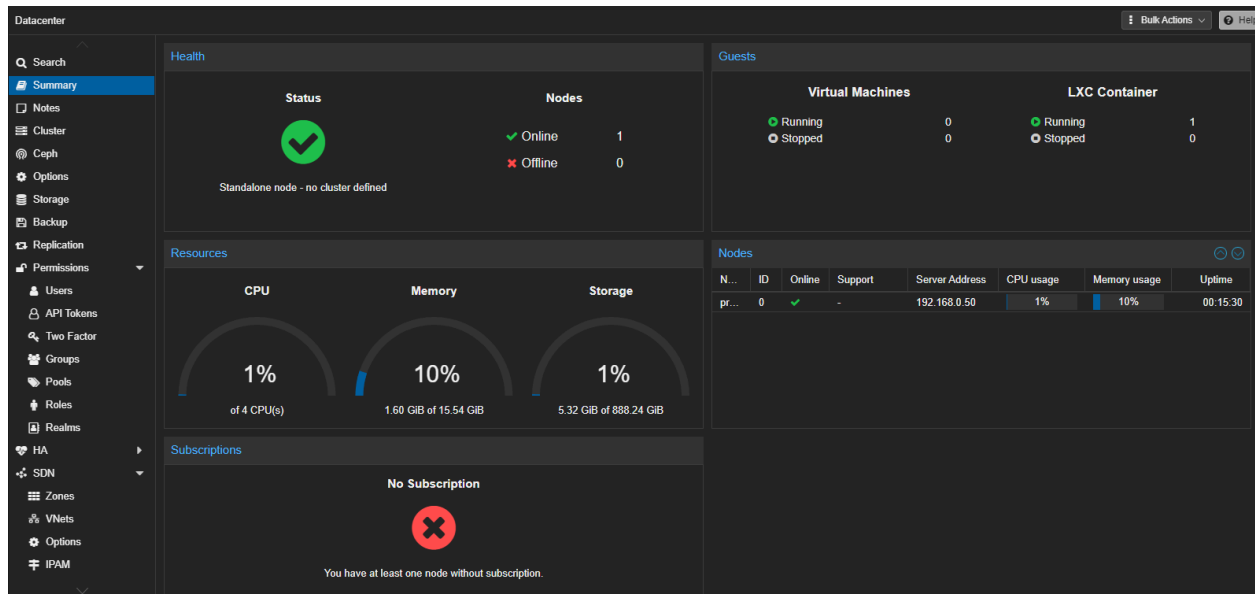
Proxmox environment overview

The host is currently running one LXC container, named pihole, on Debian 12. It uses a single storage pool built on the 1TB external SSD for now. As more VMs get added, like Wazuh and the planned Kali box, storage and RAM allocation will need to be tracked more carefully since the 16GB ceiling will be a massive constraint.

Security Posture

The host itself hasn't been hardened yet. This is next on the list, alongside setting up pfSense. Planned hardening steps include moving the web UI off its default port or restricting it to LAN only, and enabling 2FA on the root account.

Screenshots



Planned next steps

- Harden the Proxmox host itself
- pfSense VM
- Wazuh SIEM
- Kali Linux VM
- Jellyfin with GTX 960